

Mixture weight and programme strength: construct validation of single-cell topic loadings in diabetic foot skin

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Abstract

A higher mean single-cell programme score can reflect either stronger expression within cells or more cells occupying a high state. We examined this distinction before interpreting a fibroblast-associated topic as a diabetic-foot-ulcer (DFU) healing marker. A frozen 15-topic model represented 76,461 cells from 25 selected foot-skin samples. We assessed distributional semantics, technical artifacts, analysis-constant sensitivity, negative controls and external projections. An author-derived mapping collapsed 14 DFU samples to 11 patients; anatomical controls and four representation benchmarks used the corresponding analysis units. Across 25 samples, mean fibroblast loading correlated with the threshold-defined high-state fraction at $\rho = 0.933$ (sample-bootstrap 95% interval 0.789–0.983). The highest fraction, 0.841, occurred in healthy non-diabetic skin. Depth matching preserved 99.5% of state labels, and exclusion of cells carrying immune transcripts preserved sample weights ($\rho = 0.989$). Semantic associations persisted over tested thresholds and topic numbers; the healthy-skin counterexample failed at $K = 10$, and healing-arm significance was threshold-sensitive. The corrected all-cell mean-loading contrast in 11 patients was 0.0280 (permutation $p = 0.2719$; bootstrap interval $[-0.0017, 0.0631]$). The separate 9-versus-5 high-state-fraction design had 24.9% simulated power at its observed effect. Patient-level topic, module, PCA and NMF AUCs were 0.714, 0.786, 0.500 and 0.125; these exploratory estimates do not establish clinical superiority. Paired anatomy remained detectable in 10 patients. External cohorts did not supply independent patient-level healing validation. The observed loading has a substantial cell-state-abundance interpretation. Its correlation with the threshold fraction does not prove invariant within-state intensity, and the tested fibroblast pattern remains an unvalidated prognostic marker. Patient-level validation requires independently defined outcomes and a clinically justified effect margin.

Keywords: topic model; cell-state abundance; construct validity; DFU; CHI3L1; TIMP1; patient-level inference.

1 Introduction

Diabetic foot ulceration combines tissue injury, inflammation and impaired repair, and a healed wound may remain vulnerable to recurrence [1, 17]. Single-cell profiling has identified fibroblast-associated expression programmes that differ between healing groups in discovery material [14, 20]. These observations motivate prognostic research, but a tissue-level programme score must first be understood as a measurement. A higher average could reflect stronger expression within cells, a greater fraction of cells occupying a state, or both.

Topic models represent observations using mixture coordinates, while gene-expression decompositions can separate cell identity from cellular activity [3, 10]. Deep generative representations can also summarize high-dimensional expression in transferable coordinates [13]. None of these frameworks guarantees that an average over a biopsy is a cell-intrinsic intensity measurement. Relative cell abundances are compositional, so a change in one compartment can alter the apparent contribution of another [4]. Distinguishing these explanations changes the biological interpretation and the design of a subsequent validation study.

Technical robustness and statistical independence pose separate requirements. Dissociation can induce expression programmes, and ambient RNA can alter the apparent profiles of recovered cells [21, 22]. Quality control and normalization therefore matter before comparing tissues [9]. At the same time, cells from one individual share biological and sampling variation. Multi-sample analysis must preserve the specimen or patient as the appropriate independent unit; more cells do not substitute for more patients [6, 12, 18, 23].

A useful prognostic marker also requires an outcome definition, an appropriate validation population and an assessment of uncertainty that matches the intended clinical use [5]. Failure to reject an outcome difference does not establish equivalence, which requires a scientifically justified margin [11, 16]. A healthy specimen with a high score can challenge disease exclusivity without disproving a probabilistic prognostic association.

We examine state abundance, technical robustness and healing association in that order. A frozen expression map supplies cell loadings, distributional comparisons examine their measurement meaning, technical perturbations test robustness, and patient-level outcome and paired-anatomy analyses assess what the readout can distinguish. External cohorts provide additional tissue contexts with their own metadata limits. This sequence connects the biological interpretation of the measurement to the strength of the available clinical evidence.

2 Results

2.1 The frozen representation is structured, but its principal loading is a mixture coordinate

The analysis architecture is summarized in Figure 1. The frozen expression map is followed by a cell-state mixture audit, technical and environmental-RNA controls, patient-unit correction, anatomical controls and bounded external construct projections.

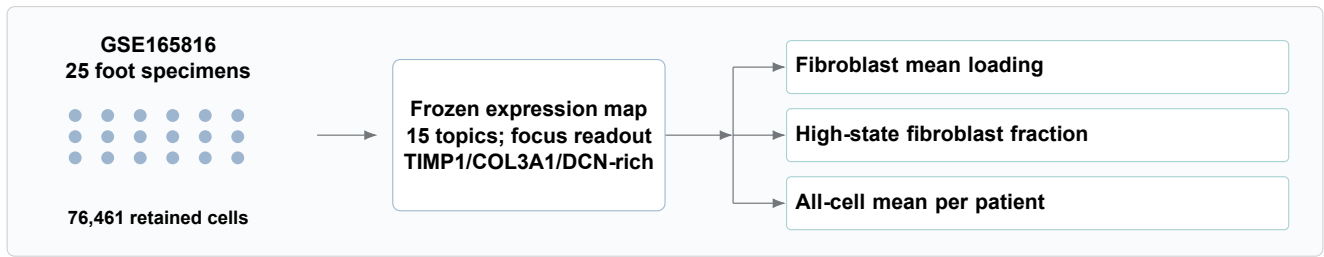
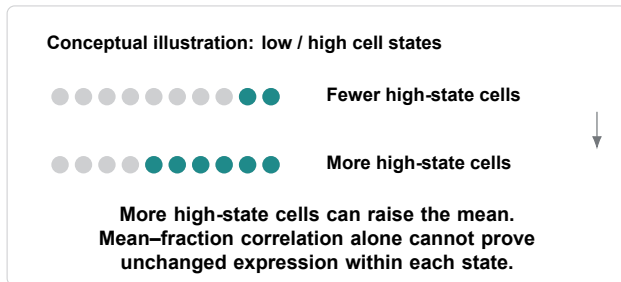
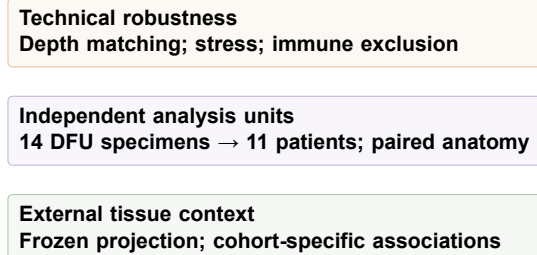
A From skin specimens to three distinct readouts**B What changes the average?****C How is the interpretation tested?****Cell-state abundance, technical robustness and healing prediction require separate evidence**

Figure 1. Study design and measurement logic. (A) The discovery series supplies 76,461 retained cells from 25 selected foot-skin specimens. A frozen 15-topic expression model produces three distinct summaries: the mean fibroblast loading, the fraction of fibroblasts above a fixed loading cut, and the all-cell mean within each mapped patient. The gene labels describe leading decoder genes, not a newly defined cell type. (B) The cell cartoons are conceptual, not observed counts: grey and teal circles denote low and high states. Changing their relative abundance can increase the sample mean without changing either state's intensity; a mean–fraction correlation alone cannot establish constant within-state expression. (C) Technical perturbations test robustness, the healing contrast collapses 14 DFU specimens into 11 patients, paired anatomy tests tissue sensitivity, and external projections assess cohort-specific construct associations. Each evidence layer addresses a different question; these analyses do not supply independent prognostic validation.

The discovery representation retained 76,461 quality-controlled cells from 25 selected foot-skin samples in the original 54-sample series. Beta perplexity was 243, and the manifold audit passed its prespecified integrity checks. Fibroblast topic was enriched for TIMP1, COL3A1, DCN, FN1, LUM and CHI3L1, with immediate-early and oxidative-stress genes also present among the top-ranked terms. The gene list therefore supports a fibroblast-associated inflammatory/stress description; it does not by itself establish a unique cell state or a healing mechanism (Figure 2).

The cell-level distribution was better described by two components than by a single continuous shift. High-loading samples retained a low-state tail overlapping the low-loading samples. Across the 25 analyzable samples, the sample mean and estimated mixture weight had Spearman $\rho = 0.933$. The bootstrap 95% interval was 0.789–0.983, and leaving out any one sample gave values from 0.924 to 0.957; the permutation null was centred near zero (Table S8). This is the central measurement result: in this dataset, the sample average is strongly associated with the high-state fraction. The two summaries are calculated from the same cells, so their correlation alone does not prove that component intensities are constant. The result does not say that high-state cells have no intrinsic biology. Together with the observed persistence of a low-state component, it supports a substantial abundance contribution to the sample-level number; it does not partition that contribution causally. Patient aggregation gave a correlation of 0.937 across 20 patients, with a 10,000-draw patient-bootstrap 95% interval of 0.776–0.990. Omitting one patient at a time gave correlations of 0.927–0.964 (Table S16), extending the specimen-resampling analysis without adding independent observations.

2.2 Technical controls support state existence, not prognostic validity

The high state survived the three artifact challenges. The dissociation stress standardized mean difference was +0.246, below the rejection threshold. PTPRC positivity was 0.98% in high-state cells and 0.62% in low-state cells, a rate ratio of 1.60. Exact panel-count matching produced 99.5% cell-label agreement and a sample-weight correlation of 0.984. The state was therefore not explained by a single depth imbalance or a simple doublet excess (Figure 3).

Removing cells carrying any of the prespecified immune transcripts left the sample weights nearly unchanged ($\rho = 0.989$) and increased the separation of the two components. This supports invariance to ambient immune RNA. The placebo analysis was deliberately retained: unrelated tissue covariates flattened residual bimodality as well as immune covariates. At 24 matched samples, the data cannot determine whether the state is statistically separable from every tissue-composition axis. The matched exclusion comparison contains 24 samples, while each residual regression uses 25 samples. The immune and placebo regressions

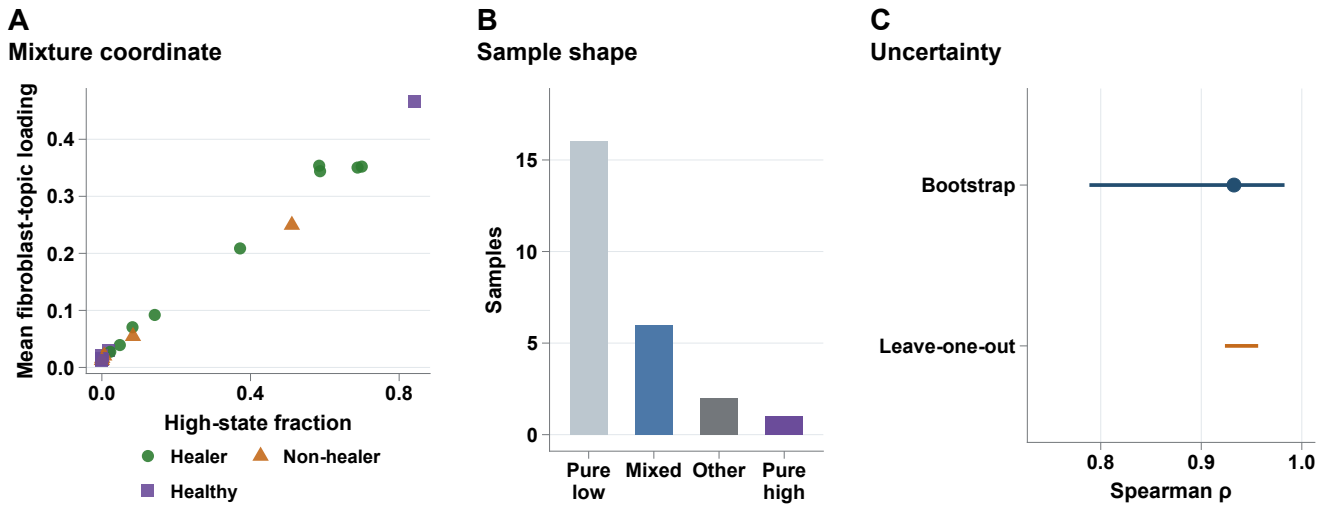


Figure 2. Association between fibroblast mean fibroblast-associated topic loading and the high-state fraction. (A) Each point is one of 25 GSE165816 specimens: 9 healing (green circles), 5 non-healing (orange triangles) and 11 healthy non-diabetic (purple squares). The high-state fraction uses the pooled cut 0.1842958; Spearman $\rho=0.933$. (B) Sample-shape counts use within-sample loading quantiles: pure low requires the 90th percentile at most 0.15, pure high the 10th percentile at least 0.10, and mixed the 10th percentile at most 0.05 and 90th percentile at least 0.40; other samples form the remaining category. (C) The dot is the observed correlation, the blue segment is the 10,000-draw sample-bootstrap 95% interval (0.789–0.983), and the orange segment is the leave-one-sample-out range (0.924–0.957). Both summaries use the same cells; the association does not by itself prove stable within-state intensity or a healing effect.

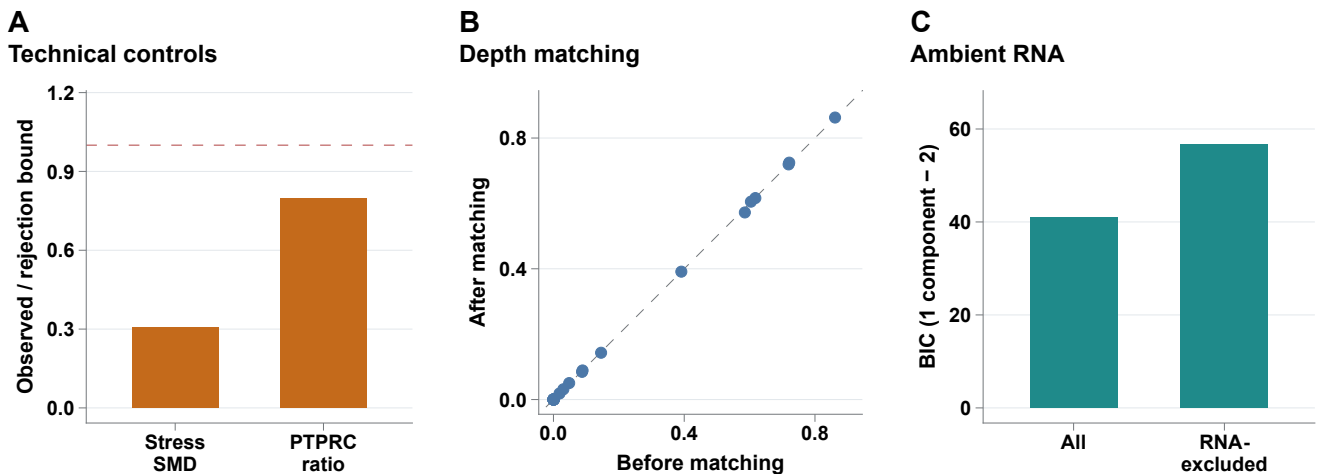


Figure 3. Technical and immune-transcript exclusion controls in the discovery cohort. (A) Dissociation standardized mean difference (SMD, 0.246) and the high-state/low-state PTPRC-positive rate ratio (1.60) are divided by rejection bounds 0.8 and 2.0; the dashed line is one. The rounded state cut is 0.184. (B) Each point is one of 25 samples after restricting to 14,561 fibroblasts eligible for panel-count matching to 986 counts per cell. Reprojection preserves sample fractions ($\rho=0.984$) and 99.5% of cell labels; the dashed line is identity. (C) Bars are BIC for one Gaussian minus BIC for two Gaussians fitted to sample high-state fractions, using 24 matched samples before and after exclusion. Positive values favour two components. Exclusion retains 11,262 of 19,410 fibroblasts overall, and matched sample fractions correlate at $\rho=0.989$. This control measures invariance to the exclusion rule; it does not identify every source of ambient RNA or tissue-composition dependence.

explained 49.5% and 35.5% of the variance, respectively, and both residual BIC differences favoured one component (Table S11). A clean technical control is not evidence that the state is a healing marker.

An exploratory scan of all 31 available covariates found 14 associations below BH-adjusted $q = 0.05$ across 25 samples. The largest positive association was with the assigned B/plasma-cell fraction ($\rho = 0.828$); CFD/APOD-rich topic mean loading was negatively associated ($\rho = -0.766$). Fibroblast PTPRC positivity and median UMI count were also associated ($\rho = 0.727$ and 0.663). The complete scan, including non-significant covariates, is in Table S10. These marginal, sample-unit correlations do not establish causal drivers or independence from tissue composition; the positive depth association remains relevant despite the successful count-matching control.

2.3 A real state is not the same as a healing marker

The high-state pattern was not restricted to samples labelled as healing. Four of nine healing samples and none of five non-healing samples were classified as high-mode samples, Fisher exact $p = 0.2208$. The most extreme high-state fraction was a healthy non-diabetic foot-skin sample, whose mixture weight was 0.841. That observation is an existence counterexample to the statement

that the state is uniquely a marker of a healing DFU. It neither identifies the tissue process responsible nor refutes a possible probabilistic association with healing.

The lineage assignment used markers disjoint from the tested topic: the marker lists shared no genes. Fibroblast composition itself differed between the healing and non-healing sample groups only descriptively (ratio 1.42, $p = 0.438$). Within the fibroblast gate, the discovery contrast was 2.89-fold, with a bootstrap 95% interval of 1.06–13.97 and sample-level $p = 0.04196$. This value is retained as a discovery observation, not as an independent patient validation, because the 14 samples represent 11 patients and include repeated same-wound, same-day specimens. The complete lineage and composition contrasts are in Table S12.

For the separate all-cell mean-loading readout, collapsing repeats to 11 patients changed the healing-minus-non-healing difference from 0.0435 to 0.0280. This is not a patient-level recalculation of the 2.89-fold fibroblast-restricted contrast. The exhaustively enumerated, add-one-corrected permutation p value was 0.2719 and the bootstrap 95% interval was $[-0.0017, 0.0631]$. The earlier 14-sample contrast was 0.0435 with $p = 0.0764$ and sensitivity margin 0.0753; the patient-unit margin was 0.0618. Neither margin is a clinical MCID. The corrected result is therefore a unit-aware discovery result with substantial uncertainty, not a definitive null test. Mapped scores are given in Table S2; equal weighting of repeated specimen means gave a similar patient contrast of 0.0284. The full equivalence sensitivity grid is in Table S15. Patient omission left the all-cell contrast positive, from 0.0139 to 0.0347 across 11 deletions (Table S16); this is an influence range, not a confidence interval or evidence of clinical discrimination.

2.4 The negative healing result has a measured power boundary

The discovery design had only 24.9% estimated power at the observed healing-minus-non-healing effect. Under the stated simulation, the minimum detectable difference at 80% power was 0.519, approximately 1.9 times the observed mean scale. Figure 4C displays this as a power boundary rather than as a null result. This calculation describes the effect size detectable with the specified design; it does not by itself exclude large or moderate healing effects (Figure 4).

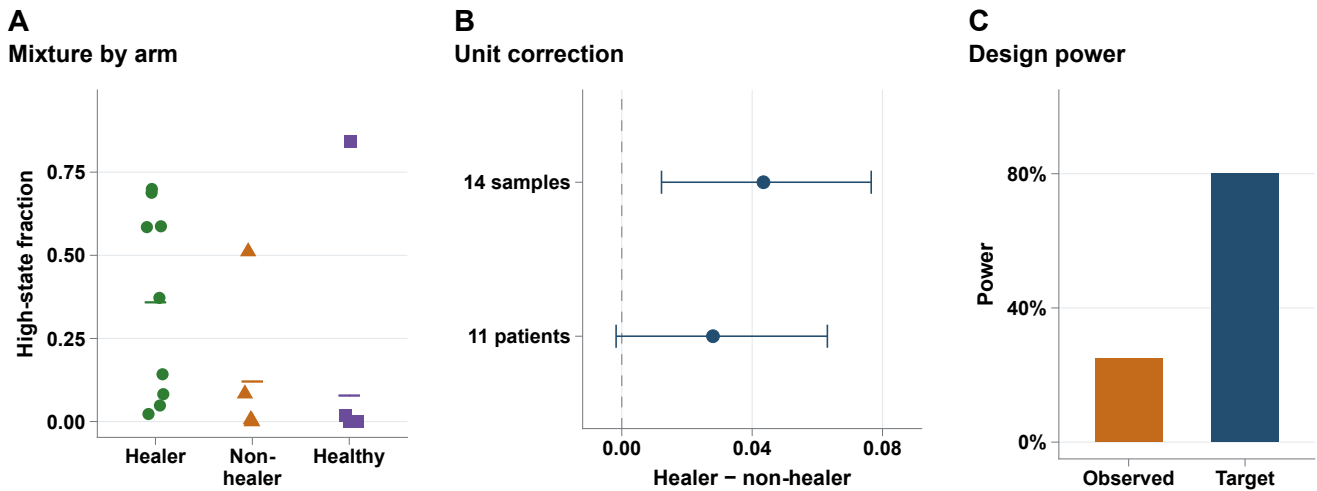


Figure 4. State abundance, analysis unit and design power. (A) Symbols show high-state fractions for 9 healing, 5 non-healing and 11 healthy specimens; horizontal segments mark arm means. Green circles, orange triangles and purple squares identify these groups. The largest fraction is 0.841 in healthy non-diabetic foot skin, showing that the state is not exclusive to healing ulcers. (B) Dots and within-arm bootstrap 95% intervals show the healing-minus-non-healing difference in all-cell mean fibroblast-associated topic loading for 14 specimens and 11 mapped patients. The patient analysis resamples 7 healing and 4 non-healing patients in 30,000 draws. Its readout differs from the fibroblast high-state fraction in A; the vertical dashed line is zero. (C) Gaussian location-shift simulation with a two-sided Mann–Whitney test estimates 24.9% power for the separate 9-versus-5 high-state-fraction design, against an 80% target. Its minimum detectable difference is 0.519. These are design sensitivities, not effect-exclusion bounds or clinical minimum important differences.

The paired anatomical control confirmed that the frozen encoder responds to a known tissue difference. Across 10 paired patients, the fibroblast-associated topic increased in foot relative to forearm skin by +0.0476 (exhaustive sign-flip $p = 0.0107$, BH $q = 0.0402$); the fibroblast-restricted contrast was +0.1398 ($p = 0.0146$). Four of 15 topics remained significant after BH correction (Table S14). The earlier 3-of-15 sample-level positive-control result is directionally consistent, but the paired analysis is the more appropriate patient-unit sensitivity check. These controls make a complete assay failure less likely, but they do not turn the underpowered DFU comparison into a proof of no association.

We also ran a design calculation for the next cohort, keeping it separate from the biological results. Under the current arm-specific spread, one observation per patient, 80% target power and no missing outcomes, a candidate difference of 0.03 would require 41 total observations, while a candidate equivalence margin of 0.05 would require 17. These values are design inputs rather than findings: the clinical MCID is still open, repeated samples increase the design effect, and the calculation cannot turn the discovery cohort into an adequately powered study. Tables S6 and S7 give the full difference, equivalence and exploratory AUC-gap design calculations.

2.5 Method constants expose both stable and fragile claims

The mean-versus-weight association remained positive across the cut grid, from $\rho = 0.909$ to 0.948 , and across $K=10, 15$ and 20 , from $\rho = 0.924$ to 0.947 . The measurement-semantic claim is therefore not an artefact of the published cut alone (Figure 5).

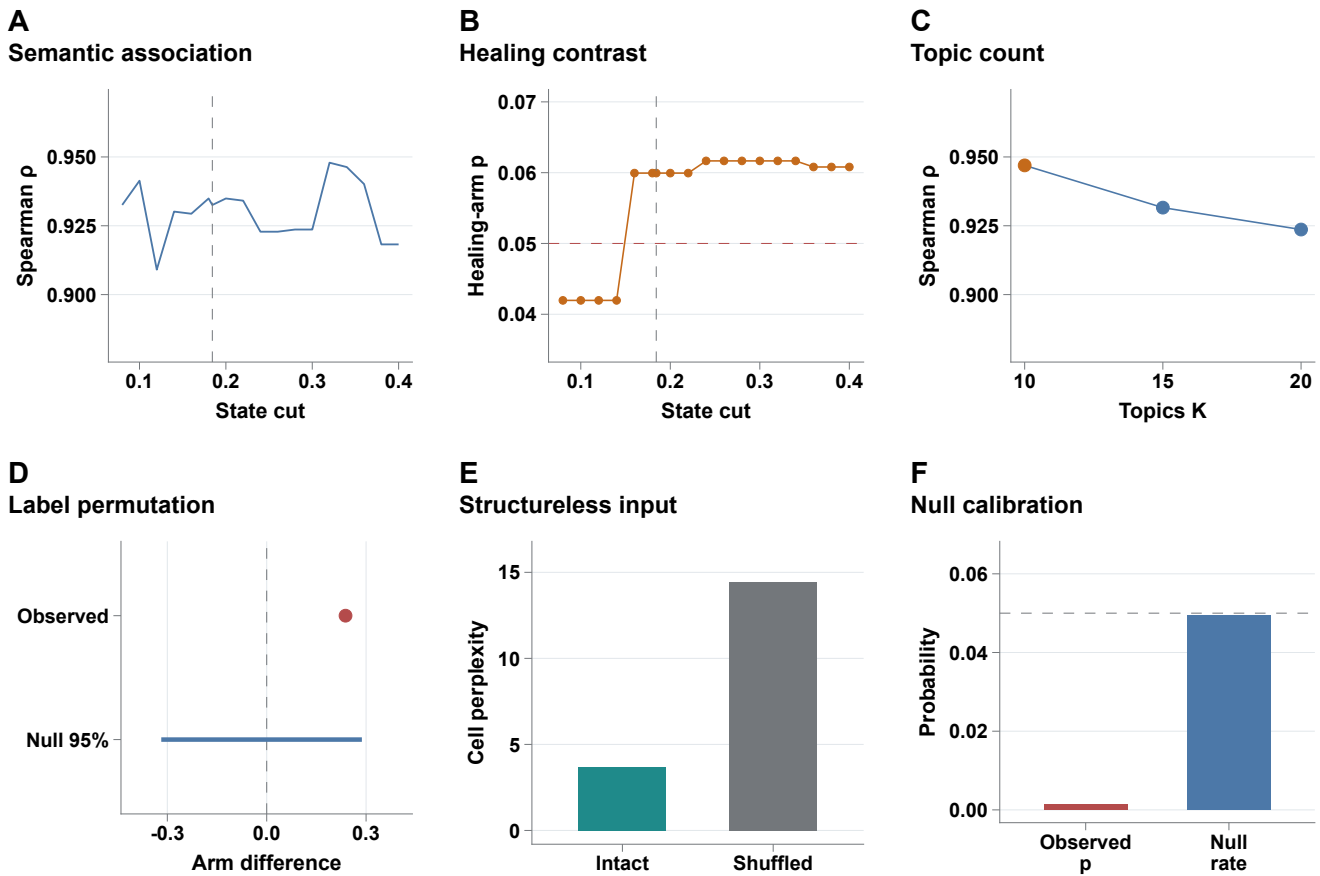


Figure 5. Threshold, capacity and negative-control analyses. (A,B) Spearman correlation of fibroblast mean loading with the high-state fraction across 25 samples and the two-sided Mann–Whitney healing-arm p value across 9 versus 5 samples, for all 18 cuts from 0.08 to 0.40. Vertical dashed lines mark the fitted cut; the horizontal line in B marks $p=0.05$. (C) Correlations for $K=10, 15$ and 20 after tracking the TIMP1-carrying topic; orange marks $K=10$, where the highest fraction is no longer in healthy skin. (D) The observed high-state-fraction arm difference is compared with the central 95% of 20,000 label permutations ($p=0.1358$). (E) Gene-column shuffling of 6,000 cells raises mean cell perplexity from 3.68 to 14.43 out of 15, violating the concentration criterion. (F) The anatomical sample-level control compares 33 foot and 11 forearm specimens: observed $p=0.0015$, with 5.0% of permuted comparisons below 0.05. This calibration is distinct from the paired 10-patient anatomy analysis and does not establish pipeline-wide error control.

The healthy-skin counterexample was more conditional. It held at all 18 tested cuts, but it failed at $K=10$ because the topic carrying TIMP1 belonged to a different IGFBP-rich programme. It held at $K=15$ and $K=20$ when the CHI3L1/COL3A1/DCN/FN1 state resolved as its own topic. The defensible wording is consequently “the counterexample holds for the tested $K=15$ and $K=20$ fits but not the $K=10$ fit,” not “the counterexample is invariant to model capacity.”

The healing-arm p value moved between 0.042 and 0.062 over the cut grid, with only 4 of 18 cuts below 0.05 (Tables S18 and S17). That sensitivity is a reason to avoid presenting the discovery p value as a stable healing signal. It also independently supports the decision to keep the healing claim unvalidated.

2.6 Negative controls and representation comparisons constrain interpretation

The label-permutation negative control produced an observed difference of +0.2380 inside the 95% null interval $[-0.3183, +0.2874]$, with $p = 0.1358$. The observed contrast was not exceptional under this label-exchangeability null. This single calibration exercise does not prove that the complete pipeline is incapable of false-positive associations. In the unstructured-input control, real data had perplexity 3.68/15, whereas gene-column shuffling produced 14.43/15 and a nearly uniform configuration that failed the integrity criterion; the destroyed co-expression structure was therefore rejected. In the permutation positive control, the observed p value was 0.0015 and the proportion of significant permutations was 5.0%, consistent with the nominal type-I error rate (Figure 5D–F).

The representation benchmark gave point AUCs of 0.822 for fibroblast-associated topic, 0.844 for the module score, 0.533 for PCA and 0.878 for NMF. No pairwise bootstrap interval for an AUC difference excluded zero. The module score alone separated its permutation null at $p = 0.0424$; fibroblast-associated topic and NMF were just above the threshold at $p = 0.0604$ and $p = 0.0597$, while PCA gave $p = 0.4527$. The null AUC means were not all 0.5 (the fibroblast-associated topic, 0.561;

NMF, 0.553), illustrating why choosing orientation inside each fold can inflate a naive 0.5 comparison. These results quantify differences without supporting a ranking (Figure 6; Tables S3, S5 and S13).

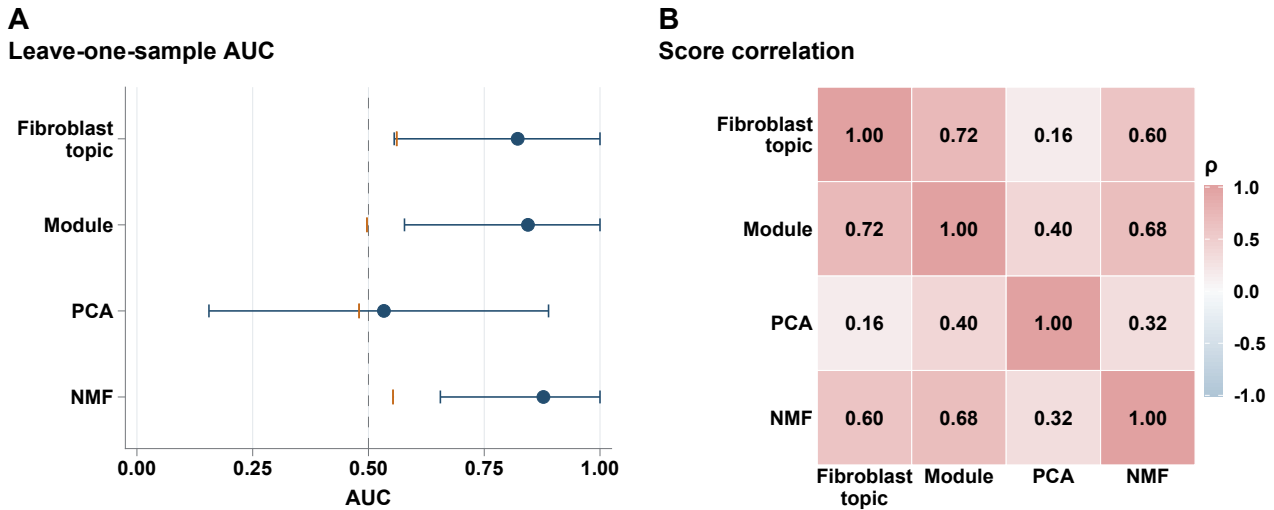


Figure 6. Sample-unit representation comparisons in 14 DFU specimens (9 healing, 5 non-healing). (A) Points are leave-one-sample-out areas under the ROC curve (AUCs); horizontal bars are percentile 95% intervals from 4,000 arm-stratified bootstrap draws of fixed out-of-fold scores, with common resampling indices across methods. Orange ticks mark the permutation-null means after rerunning outcome-dependent fitting and orientation. The dashed line at AUC=0.5 is a visual reference, not the fitted permutation null. Fibroblast topic denotes the frozen all-cell mean, Module the specified fibro-inflammatory score, PCA principal component analysis, and NMF nonnegative matrix factorization. No paired AUC-difference interval excludes zero. (B) Cells contain Spearman correlations of the 14 sample-level readouts; the colour scale and printed values encode the same statistic. These exploratory comparisons do not establish a clinical method ranking.

2.7 Patient-level readouts weaken outcome discrimination, while anatomy remains detectable

The patient-unit benchmark lowered the apparent discrimination of the frozen topic readout relative to a sample-level analysis and made the uncertainty explicit. Across 11 patients, fibroblast-associated topic AUC was 0.714 (bootstrap 95% interval 0.357–1.000; permutation $p = 0.302$), while the module score was 0.786 (0.500–1.000; $p = 0.166$). PCA was 0.500 and NMF was 0.125, with broad intervals and p values of 0.503 and 0.975. The companion sample-level score matrix shows correlations from 0.165 to 0.723 across topic, module, PCA and NMF readouts, so relatedness does not imply a stable ordering. Topic versus module and topic versus PCA differences were not separated from zero; only the topic-versus-NMF and module-versus-NMF point contrasts had bootstrap intervals excluding zero. Because this is a single, small discovery cohort, the result is a unit-sensitivity audit rather than a claim that one representation is clinically superior (Tables S4 and S13; Figure 7).

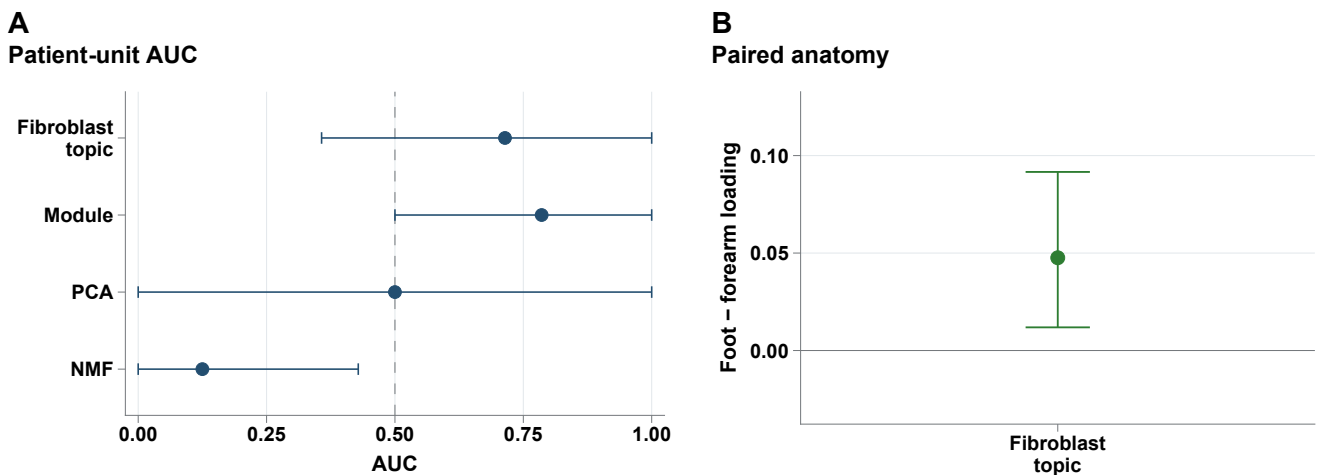


Figure 7. Patient-unit discrimination and paired anatomical sensitivity. (A) Points are leave-one-patient-out AUCs for 7 healing and 4 non-healing patients after cell-count-weighted collapse of repeated specimens. Bars are 95% intervals from 4,000 arm-stratified patient bootstrap draws of fixed out-of-fold scores; the dashed line marks AUC=0.5. Fibroblast topic, Module, PCA and NMF denote the frozen topic mean, fibro-inflammatory module, principal component and nonnegative matrix factorization readouts. (B) The point is the mean all-cell fibroblast-associated topic loading difference, foot minus forearm, across 10 paired patients; the bar is a 4,000-draw paired-patient bootstrap 95% interval and the horizontal line is zero. The exhaustive sign-flip test with add-one correction gives $p=0.0107$ and Benjamini–Hochberg $q=0.0402$ across 15 topics. Both panels use the discovery source and are not independent clinical validation.

The paired anatomical control provides the complementary positive result. The same fibroblast-associated topic readout changed between foot and forearm skin in 10 paired patients, with exact $p = 0.0107$ and BH $q = 0.0402$. The combination of weak, uncertain outcome discrimination and a detectable anatomical contrast supports a measurement-sensitivity interpretation, but it does not establish healing specificity.

2.8 External projections remain construct audits

The external immune-axis analysis used mutually exclusive lineage compartments. GSE223964 and GSE245703 gave immune-axis correlations of -0.024 and $+0.846$, respectively. Within the immune compartment, B/plasma-cell share was weakly related to the topic signal (-0.071 and $+0.077$), while correlations with median detected genes per cell were negative in both datasets (-0.429 and -0.427). The signs are inconsistent for a generalized immune-axis effect, and neither dataset has an authoritative patient-level healing endpoint (Table S9; Figure 8).

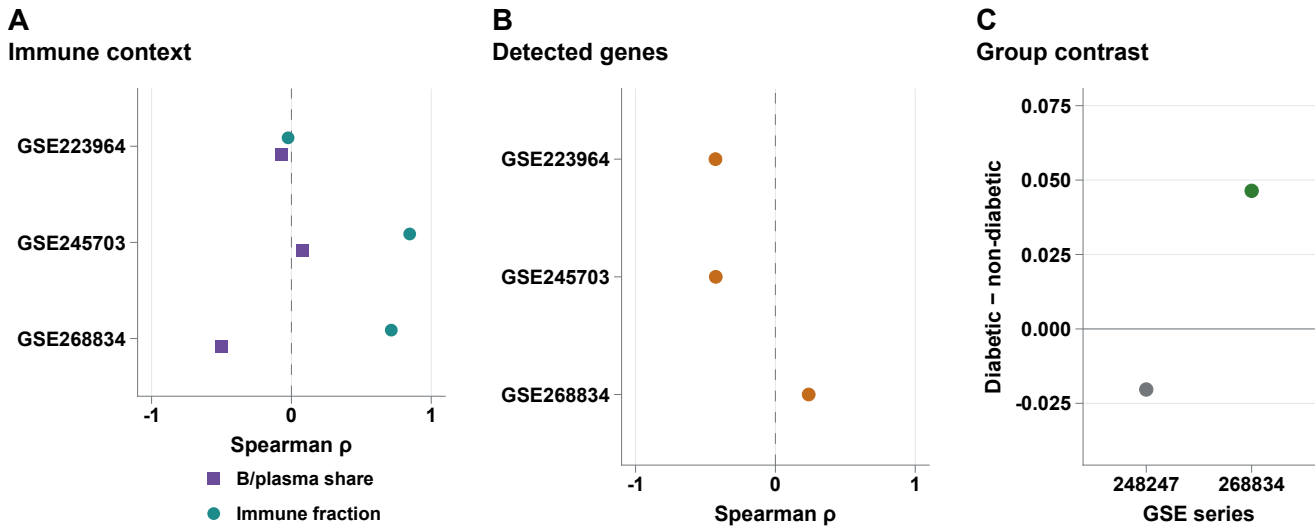


Figure 8. Frozen external projections under mutually exclusive lineage gates. (A) Each point is a cohort-level Spearman correlation between the sample fibroblast fibroblast-associated topic mean and either immune fraction among all QC cells (teal circles) or B/plasma share within immune cells (purple squares). (B) Orange points correlate the same topic mean with median detected genes per cell. GSE223964, GSE245703 and GSE268834 contribute 8, 12 and 8 specimens; vertical dashed lines mark zero. (C) Points show diabetic-minus-non-diabetic differences in the sample fibroblast mean: $+0.0464$ in GSE268834 (descriptive permutation $p=0.2812$) and -0.0204 in GSE248247 ($p=1.0$). GSE248247 contains one specimen per group and 358 QC cells, so its correlations are not estimated. No panel displays confidence intervals. These cohorts lack qualifying patient-level healing endpoints and are not pooled for prognosis inference.

GSE231643 therefore provides a weaker, label-only replication of direction: the within-fibroblast ratio falls from 2.89 in the discovery analysis to 1.66 (sample-bootstrap 95% interval 1.01–6.10; two-sided Mann–Whitney $p = 0.1429$; Table S12). The separate GSE241132 acute-wound cohort gave a broad immune-axis rho of -0.070 and an author-annotated B/plasma-cell rho of $+0.210$ across three donors; both values are descriptive and were not combined with the DFU outcome analyses. GSE268834 supplied another descriptive projection: the topic-associated fibroblast signal correlated with exclusive immune fraction at $+0.714$, and the diabetic-minus-non-diabetic fibroblast-topic contrast was $+0.0464$ with $p = 0.2812$. This is compatible with an immune/tissue-context hypothesis but cannot distinguish it from other between-sample differences.

GSE248247 was a deliberately low-yield stress test of the frozen projection. It contained 561 raw cells and 358 quality-controlled cells, with 6,994 of 7,002 panel genes covered in each sample. The fibroblast topic contrast between one diabetic and one non-diabetic sample was -0.0204 with descriptive permutation $p = 1.0$. All correlations were non-estimable because $n=2$. The exclusive compartment sum was exactly one in both samples, demonstrating that the lineage gate was non-overlapping; the dataset cannot validate healing or support pooled patient inference.

3 Discussion

The central finding concerns the interpretation of a tissue-level expression readout. Across the discovery specimens, the mean fibroblast-associated loading closely followed the fraction of cells above a fixed cut. Collapsing repeated specimens to patients preserved the association, and patient resampling supported its stability. A higher mean therefore carries substantial information about cell-state abundance. It does not by itself establish a stronger programme within each cell. The exact mean decomposition separates the abundance and within-state terms, but their empirical correlation does not identify a causal contribution or prove constant state-specific intensity.

This distinction complements expression-programme decomposition and compositional analysis [4, 10]. An expression programme can be reproducible while its biopsy-level average changes with the sampled cellular mixture. Gene labels describe the fitted loading pattern; they do not establish a new cell type or an exclusive mechanism of repair. Independent measurements of state-specific expression and tissue abundance would be needed to resolve the remaining biological ambiguity.

The technical controls make several simple explanations less plausible. State labels survived count matching, and immune-transcript exclusion preserved specimen summaries. Dissociation and ambient contamination are established concerns in single-cell analysis [21, 22], so these checks support interpreting the pattern as more than one obvious processing artifact. The placebo residual analysis also gives a useful limit: multiple tissue-composition covariates reduce residual bimodality. The current data cannot uniquely separate the loading from every compositional influence.

Clinical interpretation remains more restricted. The strongest high-state fraction occurred in healthy skin, the healing comparison was imprecise after patient aggregation, and the apparent result depended on the threshold used to summarize cells. Patient omission preserved the direction of the small all-cell contrast, which argues against a single-patient sign reversal but does not establish prognostic performance. The contrast between detectable paired anatomy and uncertain healing discrimination indicates sensitivity to tissue differences without establishing healing specificity. These findings agree with the need to preserve biological replication and to separate exploratory associations from prediction-model validation [5, 18, 23].

The external projections extend tissue context rather than supplying a common validation endpoint. Cohorts differ in sampling, cell recovery and available labels; some lack patient mappings or healing outcomes, and the smallest projection has only two specimens. The inconsistent immune associations suggest that tissue context deserves direct measurement in a future study. Pooling these cohorts as if they measured the same patient-level prognosis would obscure their design differences.

Several limitations constrain the conclusions. The outcome cohort contains only 11 patients and the paired-anatomy control comes from the same discovery source. The frozen encoder was trained on the expression cohort subsequently used for the internal discrimination audit. State definitions are data-derived, and threshold and topic-capacity sensitivity remain relevant. The source outcome is a 12-week category, not an individual healing time. Power and equivalence calculations quantify sensitivity under stated assumptions; a nonsignificant test or a data-derived margin cannot establish a clinical null [11]. These limitations motivate a prospective study with an independently specified patient outcome, a clinically justified effect margin, standardized tissue sampling and a frozen analysis. Such a design would test whether a measurement with substantial abundance semantics adds useful prognostic information.

4 Methods

4.1 Cohort selection and independent units

GSE165816 comprises 54 original samples and 27 author-mapped participants [14, 20]. The frozen model used 25 selected foot-skin samples in the non-diabetic, DFU-healing and DFU-non-healing groups: 81,602 cells before cell quality control and 76,461 afterwards. Cells required at least 200 detected genes and a mitochondrial count fraction at most 0.20; genes required detection in at least 10 cells before panel selection. The final panel contained 7,002 genes, obtained from a nominal 7,000-gene dispersion selection with retained anchor genes. The discovery outcome subset contained 9 healing and 5 non-healing samples.

The clinical supplement maps those 14 outcome samples to 11 patients: 7 healing and 4 non-healing. G4/G23, G7/G8 and G33/G34 are repeated samples from the same wound on the same day. We combine each patient’s retained cells before computing the primary patient score. The outcome is the study’s healing category within a 12-week follow-up window, not a measured individual healing time. Series metadata describe a shared Emory 10x Chromium 3’ V2/V3 workflow and NovaSeq instrumentation; the absence of per-sample chemistry and flow-cell identifiers prevents run-specific batch adjustment.

4.2 What the frozen neural topic model computes

For nonnegative panel counts x_{ig} with positive total, let $u_{ig} = x_{ig} / \sum_{h=1}^G x_{ih}$. The implemented model uses a diagonal-Gaussian encoder in unconstrained coordinates and a softmax decoder:

$$q_{\phi}(z | u_i) = \mathcal{N}(\mu_{\phi}(u_i), \text{diag } \sigma_{\phi}^2(u_i)), \quad (1)$$

$$\theta_{ik} = \frac{e^{z_{ik}}}{\sum_{\ell=0}^{K-1} e^{z_{i\ell}}}, \quad \beta_{kg} = \frac{e^{b_{kg}}}{\sum_{h=1}^G e^{b_{kh}}}, \quad \hat{u}_{ig} = \sum_{k=0}^{K-1} \theta_{ik} \beta_{kg}. \quad (2)$$

Equations (1)–(2) define the encoder, simplex coordinates and normalized topic-gene decoder used below. Thus $\theta_i \in \Delta^{K-1} = \{a \in \mathbb{R}^K : a_k \geq 0, \sum_{k=0}^{K-1} a_k = 1\}$ and every topic-gene row sums to one. The topic indices start at zero. This is a logistic-normal neural topic model; it does not implement an exact Dirichlet posterior or a Dirichlet prior, despite the general topic-model motivation [3, 19]. Training minimizes

$$\mathcal{L}(\phi, b) = \frac{1}{N} \sum_i \mathbb{E}_{q_{\phi}} \left[- \sum_g u_{ig} \log(\hat{u}_{ig} + 10^{-10}) \right] + 0.01 \frac{1}{N} \sum_i \text{KL}(q_{\phi}(z | u_i) \| \mathcal{N}(0, I)). \quad (3)$$

Equation (3) is the fitted objective: the reconstruction term is cross-entropy on per-cell proportions and the second term is the stated Gaussian regularizer; it is not a depth-weighted count likelihood. The encoder has widths 256 and 128, batch normalization, ReLU activations and dropout 0.1. The recorded fit used $K = 15$, 40,000 Adam steps, learning rate 10^{-3} and batches of 512. Deterministic projection uses $z = \mu_{\phi}(u)$ and $\theta = \text{softmax}(\mu_{\phi}(u))$, which is not generally $\mathbb{E}_q[\text{softmax}(z)]$. Missing panel genes in external projections are set to zero. Normalization and quality control can alter biological interpretations, so this definition is kept distinct from other variance-stabilizing methods [8, 9].

Topic-gene perplexity is summarized as $\exp\{-K^{-1} \sum_{k,g} \beta_{kg} \log \beta_{kg}\}$; it measures concentration of decoder weights, not superiority over other representations. The latent coordinate $\mu \in \mathbb{R}^K$ is not the simplex coordinate θ . No downstream outcome comparison substitutes one for the other.

4.3 Threshold fractions and the exact mean decomposition

Let $Y_{si} = \theta_{si,*}$ for fibroblast-gated cell i in sample s . Lineage markers are disjoint from the tested topic-gene list by an explicit overlap check; the gate is nevertheless an expression-based operational definition, not an independently measured cell identity. A pooled two-Gaussian fit provides a descriptive cut $c = 0.1842958$, the midpoint of the fitted component means. Gaussian components approximate a bounded score distribution and are not assumed to be literal biological populations.

$$\bar{Y}_s = \frac{1}{n_s} \sum_i Y_{si}, \quad \hat{w}_s(c) = \frac{1}{n_s} \sum_i \mathbf{1}\{Y_{si} > c\}. \quad (4)$$

Equation (4) defines both the sample mean and the reported hard-threshold fraction; the latter is not an average posterior responsibility of the Gaussian mixture. Define $\bar{Y}_{s,L}$ and $\bar{Y}_{s,H}$ as means below/at and above the cut. Then, with empty-component contributions interpreted as zero,

$$\bar{Y}_s = (1 - \hat{w}_s) \bar{Y}_{s,L} + \hat{w}_s \bar{Y}_{s,H}. \quad (5)$$

Equation (5) is exact. Only under an additional approximate component-stability assumption, $\bar{Y}_{s,L} \approx m_L$ and $\bar{Y}_{s,H} \approx m_H$, does it reduce to $\bar{Y}_s \approx m_L + (m_H - m_L) \hat{w}_s$. Consequently, a large correlation between the two summaries cannot independently prove stable intensity: both summaries use the same cells. Persistence of overlapping low-state tails supplies complementary distributional evidence. The 10th percentile of a high-loading sample (0.029) overlaps the scale of the 90th percentile in a low-loading sample (0.119); this observation alone is not a formal test of multimodality.

The semantic statistic is $\rho_{\text{sem}} = \text{cor}_{\text{Spearman}}(\bar{Y}_s, \hat{w}_s)$ across 25 samples. A 10,000-draw sample-row percentile bootstrap, 10,000 row permutations and leave-one-sample-out recalculation characterize its stability. These do not create independent patients. The patient-collapsed correlation of 0.937 across 20 mapped patients is an additional descriptive unit check.

A separate patient-resampling analysis collapses the 25 specimen means and high-state fractions to 20 patients using fibroblast counts as within-patient weights. It draws 10,000 patient samples with replacement (seed 217) and reports the percentile interval for their Spearman correlation. Leave-one-patient-out analyses recompute this correlation across the remaining 19 patients and separately recompute the all-cell healing contrast across the remaining 10 outcome patients. The reported omission range is a sensitivity diagnostic, not an interval estimate (Table S16).

4.4 Technical controls and constant sensitivity

Dissociation-induced programmes and ambient transcripts motivate the technical controls [21, 22]. The stress comparison uses a standardized mean difference (SMD) with rejection bound 0.8; the doublet-oriented control compares PTPRC-positive rates with ratio bound 2.0. Exact panel-count matching reprojects downsampled counts, comparing cell labels and sample fractions. Removing any cell carrying a specified immune transcript tests invariance to that exclusion rule. This cannot by itself identify ambient contamination or exclude every doublet. Residual mixture checks with immune covariates are compared with four unrelated placebo covariates; similarly flattened residuals limit claims of statistical separability.

Sensitivity fits use 18 cuts between 0.08 and 0.40 (including the rounded fitted cut) and $K \in \{10, 15, 20\}$. The TIMP1-carrying topic is tracked descriptively across capacities; topic labels are not assumed exchangeable or biologically identical across K . Label permutation uses 20,000 shuffles. Gene-column shuffling preserves each gene's marginal counts while destroying its joint cellwise association with other genes. The resulting topic concentration and integrity statistics are compared with real input. The anatomical label-permutation calibration checks a finite empirical false-positive fraction; it is not a proof of pipeline-wide error control. These analyses are retrospective computational controls, not a registered clinical trial.

The exploratory covariate scan retains the original rounded rule $Y_{si} \geq 0.184$, used by the technical-control analyses, rather than substituting the fitted cut 0.1842958. It correlates the sample high-state fraction with 14 other fibroblast topic means, 11 whole-sample lineage fractions, total and fibroblast cell counts, fibroblast dissociation score, median UMI count, PTPRC-positive fraction and the numeric G specimen label. Two-sided Spearman probabilities are adjusted across all 31 comparisons by Benjamini-Hochberg. The 25 rows are specimens, include repeated patients, and receive no clinical-outcome adjustment. The G label is an identifier covariate, not a verified sequencing batch or collection time. Topic coordinates and lineage fractions have compositional dependence.

4.5 Patient contrast, permutation and equivalence sensitivity

Let $\mathcal{C}_p$ contain all retained cells in the mapped DFU foot specimens of patient p . The primary corrected score is $S_p = |\mathcal{C}_p|^{-1} \sum_{i \in \mathcal{C}_p} \theta_{i,*}$, so repeated specimens contribute in proportion to retained cell counts. This all-cell readout differs from $\bar{Y}_s$ restricted to fibroblasts and from $\hat{w}_s$. For healing set H and non-healing set N ,

$$\hat{\Delta} = \frac{1}{|H|} \sum_{p \in H} S_p - \frac{1}{|N|} \sum_{p \in N} S_p. \quad (6)$$

Equation (6) is the primary patient-unit contrast; it is separate from the sample-level fibroblast ratio and from the threshold fraction. All $\binom{11}{7} = 330$ patient-label allocations are enumerated. To reproduce the recorded result, the implementation retains an add-one correction even for exhaustive enumeration:

$$p_+ = \frac{1 + \sum_{\pi \in \Pi} \mathbf{1}\{|\widehat{\Delta}_\pi| \geq |\widehat{\Delta}|\}}{1 + |\Pi|}. \quad (7)$$

Equation (7) is the recorded add-one-corrected two-sided permutation probability. It is slightly conservative relative to the uncorrected exhaustive fraction; the add-one convention is usually motivated for randomly drawn permutations [15], whereas all allocations are enumerated here. Exchangeability is an assumption for an observational label comparison, not a randomization supplied by study design. A 30,000-draw bootstrap resamples patients within outcome arms [7]. Small group sizes and conditional resampling explain why bootstrap intervals and permutation tests need not be dual.

For a symmetric equivalence margin M , the Welch two-one-sided-test procedure [11, 16] uses $SE = (s_H^2/n_H + s_N^2/n_N)^{1/2}$ and Welch degrees of freedom ν :

$$p_L = 1 - F_{t_\nu}\left(\frac{\widehat{\Delta} + M}{SE}\right), \quad p_U = F_{t_\nu}\left(\frac{\widehat{\Delta} - M}{SE}\right). \quad (8)$$

Equation (8) gives the two one-sided tests used for a prespecified margin. Both must be below 0.05 to reject non-equivalence. The data-derived infimum $M_* = |\widehat{\Delta}| + t_{\nu,0.95}SE$ is reported only as sensitivity (0.0618 at the patient unit; 0.0753 at the original sample unit). Selecting this margin after observing the data is not a clinically justified equivalence claim. No clinical minimum important difference has been supplied.

4.6 Discrimination, anatomy and design simulations

For held-out prediction scores r , AUC is the empirical probability that a healing score exceeds a non-healing score, with half credit for ties:

$$\widehat{AUC} = \frac{1}{n_H n_N} \sum_{h \in H, n \in N} \left[\mathbf{1}\{r_h > r_n\} + \frac{1}{2} \mathbf{1}\{r_h = r_n\} \right]. \quad (9)$$

Equation (9) is the pairwise empirical AUC used for the held-out scores. Topic orientation, module preprocessing and the choice of PCA/NMF components are determined inside each training fold. The topic encoder itself was trained without outcomes on the discovery cohort, including held-out patients' expression: this is an internal discrimination audit of a frozen discovery representation, not fully external representation validation. Bootstrap intervals resample fixed out-of-fold scores with common arm-stratified indices across methods; the entire fitting/orientation process is rerun for label permutations. All allocations are used for topic and module readouts; PCA and NMF use 200 sampled relabellings. Sample-level comparisons use 14 specimens and patient-level comparisons use 11 mapped patients; they are not interchangeable.

The paired anatomical control tests foot-minus-forearm differences in 10 mapped patients using all 2^{10} sign allocations and the same add-one convention. Across 15 topics, Benjamini–Hochberg adjustment is $q_{(i)} = \min\{1, \min_{j \geq i} 15p_{(j)}/j\}$, with the usual assumptions required for false-discovery-rate interpretation [2].

The historical high-state-fraction power calculation uses 9 versus 5 specimens, Gaussian location shifts with pooled SD 0.283 and a two-sided Mann–Whitney test. Its 24.9% power and $MDE_{0.8} = 0.519$ apply to this approximate design, not to the 11-patient all-cell contrast. Gaussian simulation can leave the unit interval, so it is a design-scale sensitivity approximation. A separate next-cohort simulation uses arm-specific SDs of all-cell mean scores, 0.04845 and 0.01584, Welch testing or Welch equivalence testing, and 4,000 draws per candidate size. The candidate totals are approximate conditional calculations, not validated recruitment targets (Table S6).

4.7 External projections and limits of identification

Every external cohort is projected into the frozen panel/model without outcome-guided retuning. The immune and B/plasma gates exclude promiscuous markers such as CD74, CD37, CD38 and SDC1 and assign non-overlapping compartments. Immune fraction uses all retained cells as its denominator; B/plasma share within immune cells uses only that compartment. Their correlations estimate different quantities. Negative depth correlations cannot by themselves exclude depth confounding, because a marginal sign is not a conditional adjustment.

GSE231643 supplies eight title-labelled specimens but lacks an authoritative patient mapping. GSE241132 contributes an acute-wound, three-donor construct summary. GSE223964, GSE245703 and GSE268834 lack qualifying patient-level healing endpoints, and GSE248247 has only two samples. We do not pool these correlations or describe them as independent prognostic validation. No analysis identifies a treatment effect, a time-to-healing model or a causal effect of immune infiltration.

Data and reproducibility

The discovery series is GSE165816. External projections use GSE231643, GSE241132, GSE223964, GSE245703, GSE268834 and GSE248247; each accession is linked in Table S1. This is a secondary analysis of publicly released data, with no new recruitment or intervention. The original studies govern participant consent and source-study ethical approvals. Shared series-level laboratory metadata do not identify sample-specific sequencing batches. The mathematical definitions, analysis units, numerical

summaries and figure source data specify the present analysis; no unavailable patient identifiers, healing times or independent clinical validation are imputed. Analysis code is released under the MIT License. Manuscripts, figures and source-study data remain under their respective rights and repository terms. The accompanying software archive, authored by Zeyu Fu, is available at <https://github.com/PeterPonyu/single-cell-wound-analysis>.

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Supplementary methods and numerical results

S1 Cohort roles and evidence units

Table S1 fixes the role of each accession. Specimen counts, donor counts and cell counts answer different questions. The 54-sample source series is not the 25-sample frozen foot-skin subset, and neither is the 11-patient outcome subset. Cells in an external specimen are not independent replications of that specimen’s clinical label.

Table S1. Cohort roles, analysis units and interpretation limits. GEO accessions identify the public source studies. Counts refer to specimens, mapped patients or donors as explicitly stated, rather than independent cells. DFU denotes diabetic foot ulcer and QC denotes quality control. The original discovery series, selected foot-skin representation and outcome subset have different denominators; external projections test construct associations or feasibility without a qualifying patient-level healing validation.

Cohort	Units	Role and limit
GSE165816	54 original specimens; 25 selected foot specimens	Discovery representation; 14 DFU specimens map to 11 outcome patients. The original series maps to 27 participants.
GSE231643	8 specimens	Title-derived healing labels (5, 3); no authoritative independent patient mapping.
GSE241132	3 donors; 12 timepoint samples	Healthy acute-wound construct audit; no DFU prognosis inference.
GSE223964	8 specimens	Frozen projection and exclusive immune-compartment audit; no qualifying healing outcome.
GSE245703	12 specimens	Frozen projection and immune-compartment audit; no qualifying healing outcome.
GSE268834	8 specimens	Descriptive diabetes-group and immune-context contrasts, not healing validation.
GSE248247	2 specimens; 358 QC cells	Low-yield projection check from 561 raw cells; correlations not estimated.

S2 Operational lineage and mixture rules

Discovery lineage scores average gene-wise standardized $\log(1 + 10^4 x_{ig} / \sum_h x_{ih})$ values within marker sets; the highest lineage score assigns the label. The fibroblast markers are COL1A1, COL1A2, PDGFRA, TWIST2, COL6A3, MMP2 and CCDC80. The circularity check concerns overlap with the tested topic’s specified top-gene list; it does not imply independence from all genes contributing to the neural representation. The loading is a learned function of the full panel.

In external audits, immune assignment is PTPRC detection or detection of at least two markers within a T, NK, myeloid, B or plasma panel. These panels respectively contain CD3D, CD3E, CD2, TRBC1, TRBC2, NKG7, GNLY, LYZ, CD68, CD14, ITGAX, FCER1G, TYROBP, AIF1; MS4A1, CD79A, CD79B, BANK1; and MZB1, JCHAIN, DERL3, TNFRSF17. Fibroblast-like cells require at least two of COL1A1, COL1A2, COL3A1, DCN, LUM, PDGFRA, THY1 and no immune assignment. B/plasma cells are immune-assigned cells detecting at least one B or plasma marker. All remaining cells form the other compartment. Unlike the discovery lineage list, this external structural list includes topic-associated genes; it is a descriptive operational gate and is not evidence of gene-disjoint external validation.

The disjoint immune, fibroblast-like and other compartments sum to one. B/plasma is a subset of immune, so its within-immune share must not be added as a fourth exclusive compartment. The older overlapping marker-detection fractions are not used for the current external associations.

For distributional descriptions, pure high requires the sample’s fibroblast 10th percentile to be at least 0.10; pure low requires its 90th percentile to be at most 0.15; mixed requires a 10th percentile at most 0.05 and a 90th percentile at least 0.40. Remaining samples are labelled other. These operational categories do not prove distinct biological attractors.

S3 Representation integrity is a restricted diagnostic

Let the reference and candidate k -nearest-neighbour graphs have component labels r_i, c_i . If n_{ab} counts cells jointly in reference component a and candidate component b , and $n_a = \sum_b n_{ab}$, the implementation computes reachability preservation exactly as

$$R_{\text{reach}} = \frac{\sum_{a,b} n_{ab}^2 - N}{\sum_a n_a^2 - N}, \quad (\text{S1})$$

Equation (S1) is assigned value one when the denominator is zero. This sparse component-count calculation avoids materializing all pairwise reachability values. It is a connectivity statistic: two connected graphs can give one despite severely distorted local geometry. The separate edge-survival score is reported but is not included in the binary integrity condition.

Mean cell perplexity is $\bar{P} = N^{-1} \sum_i \exp[-\sum_k \theta_{ik} \log \theta_{ik}]$. The recorded integrity condition is $R_{\text{reach}} > 0.70$ and $\bar{P} \leq 0.95K$. Gene-column shuffling raises $\bar{P}$ from 3.68 of 15 to 14.43 of 15 and fails the concentration criterion despite retained connectivity. This rejects the tested near-uniform input; it does not validate every topological feature or every cut at the fixed reachability bound. The K and state-cut grids in the main text do not constitute a sensitivity scan of the 0.70 connectivity bound.

S4 Resampling and future-study assumptions

Every interval names its resampling unit. The semantic bootstrap resamples sample rows. The all-cell outcome bootstrap resamples patients within arms after replicate collapse. AUC intervals resample fixed out-of-fold predictions; they do not refit the encoder or reproduce the entire learning algorithm on each bootstrap draw. Permutation inference reruns outcome-dependent score orientation and representation selection, which is why the appropriate null AUC need not be 0.5. The unadjusted $p = 0.0424$ for the sample-level module is exploratory across several readouts and is not a confirmatory multiplicity-controlled endpoint.

The next-study design calculation uses sample-derived arm spreads. This creates uncertainty not reflected by its Monte Carlo standard error; at estimated power 0.8 and 4,000 simulations, the conditional standard error is about $\sqrt{0.8(0.2)/4000} = 0.0063$. Candidate equivalence designs assume a true difference of zero. Bounded topic scores are approximated by Gaussian locations and variances, and proposed totals near 80% power are approximate search outputs rather than guaranteed minima.

For the exploratory AUC-design simulator, a unit-variance binormal score with class-mean separation d has $\text{AUC} = \Phi(d/\sqrt{2})$, hence $d = \sqrt{2}\Phi^{-1}(\text{AUC})$. Correlation is imposed on within-class noise, not on the full mean-shifted scores. Its standard-error approximation is estimated across simulated AUC differences; it is a model-based planning exercise, not a validated patient-level test. Neither these AUC calculations nor an inpatient design-effect formula licenses treating repeated biopsies as independent prognosis cases.

S5 Numerical results

Tables S2–S9 present the mapped patient scores, discrimination intervals, cross-method associations, conditional design calculations, semantic uncertainty and external immune-context summaries. Their denominators and limits accompany the values so that the three distinct readouts—all-cell mean, fibroblast mean and high-state fraction—remain identifiable. Tables S10 and S11 retain the complete covariate screen and immune/placebo residual summaries. Tables S12, S13, S14 and S15 distinguish the lineage, discrimination, paired-anatomy and equivalence estimands. Tables S18 and S17 retain every tested threshold and topic capacity, including adverse sensitivities. Table S7 records the exploratory AUC-gap simulation. The separate arithmetic conversion from specimens to patients under assumed inpatient correlation is not a validated patient-prognosis design and is not used as a recruitment recommendation.

Table S2. Author-mapped GSE165816 DFU patient scores (7 healed, 4 not healed). Each score is the mean frozen fibroblast-associated topic loading over all retained cells in the listed foot specimens; pooling cells weights repeated specimens by retained cell count. Cells are counts, not independent outcome observations. Patient identifiers are study-derived pseudonyms and G labels identify specimens. The outcome is the source-study category within 12 weeks, not an individual healing day. Scores are on the topic-loading scale from zero to one.

Patient	Outcome	Specimens	Cells	All-cell mean
S01	not healed	G6	2279	0.0500
S02	healed	G23, G4	7495	0.1448
S03	healed	G2	3951	0.0213
S04	healed	G15	2250	0.0749
S05	not healed	G9	4290	0.0364
S06	not healed	G33, G34	7681	0.0131
S07	not healed	G39	2481	0.0305
S08	healed	G7, G8	4192	0.0751
S09	healed	G49	2928	0.0539
S10	healed	G42	1838	0.0201
S11	healed	G45	3374	0.0331

Table S3. Exploratory discrimination across 9 healing and 5 non-healing GSE165816 specimens. AUC is the area under the ROC curve for leave-one-sample-out scores; higher values favour healing. The percentile 95% intervals use 4,000 arm-stratified resamples of fixed out-of-fold scores. One-sided permutation probabilities and null means rerun fold-dependent orientation and fitting. PCA is principal component analysis; NMF is nonnegative matrix factorization. The frozen topic encoder itself used the discovery expression cohort. Repeated patients limit independence, and none of the six paired AUC-difference intervals excludes zero.

Readout	AUC	Bootstrap 95% interval	Permutation p	Null mean
Fibroblast topic	0.822	0.556 to 1.000	0.0604	0.561
Module	0.844	0.578 to 1.000	0.0424	0.497
PCA	0.533	0.156 to 0.889	0.4527	0.480
NMF	0.878	0.655 to 1.000	0.0597	0.553

Table S4. Patient-unit discrimination and anatomical sensitivity within GSE165816. The first four rows are leave-one-patient-out AUCs for 7 healing and 4 non-healing patients, with 4,000 arm-stratified bootstrap draws of fixed scores and one-sided label-permutation probabilities. PCA and NMF denote principal component analysis and nonnegative matrix factorization. The final row is the foot-minus-forearm all-cell fibroblast-associated topic mean difference in 10 paired patients, with a 4,000-draw paired bootstrap interval and a two-sided exhaustive sign-flip probability using add-one correction. Its Benjamini–Hochberg q value across 15 topics is 0.0402. The two row types have different estimands; neither supplies external validation.

Readout	Estimate	Bootstrap 95% interval	Permutation p
Fibroblast topic	0.7143	0.3571 to 1.0000	0.3021
Module	0.7857	0.5000 to 1.0000	0.1662
PCA	0.5000	0.0000 to 1.0000	0.5025
NMF	0.1250	0.0000 to 0.4286	0.9751
foot minus forearm	0.0476	0.0119 to 0.0916	0.0107

Table S5. Spearman correlations between the four representative sample-level prediction scores across the same 14 DFU specimens (9 healing, 5 non-healing). Fibroblast topic is the frozen all-cell loading, Module the specified fibro-inflammatory score, PCA principal component analysis and NMF nonnegative matrix factorization. Each row is one pair of readouts; values are descriptive point estimates without confidence intervals or multiplicity-adjusted tests. Related scores do not establish equivalent discrimination or rank methods.

Readout A	Readout B	Spearman correlation
Fibroblast topic	Module	0.7231
Fibroblast topic	PCA	0.1648
Fibroblast topic	NMF	0.6023
Module	PCA	0.4022
Module	NMF	0.6756
PCA	NMF	0.3200

Table S6. Conditional next-cohort simulations for all-cell fibroblast-associated topic mean loading. Rows assume one independent patient per observation, no missing outcomes, healing allocation near 9/14, and arm SDs 0.04845 and 0.01584 estimated from discovery specimens. Two-sided Welch tests assess candidate mean differences; Welch two-one-sided tests assess equivalence under a true difference of zero, at $\alpha=0.05$. Total and arm columns give the simulated observation counts; power uses 4,000 draws per candidate size. Values near 0.80 have Monte Carlo error and are approximate search outputs. Candidate differences and margins are inputs, not clinical minimum important differences or validated recruitment targets.

Design	Difference / margin	Total	Healing / other	Estimated power
Detect difference	0.02	87	56 / 31	0.802
Detect difference	0.03	41	26 / 15	0.798
Detect difference	0.0435	21	14 / 7	0.822
Detect difference	0.06	12	8 / 4	0.807
Detect difference	0.08	9	6 / 3	0.862
Detect difference	0.1	7	4 / 3	0.786
Equivalence under zero difference	0.02	94	60 / 34	0.805
Equivalence under zero difference	0.03	43	28 / 15	0.803
Equivalence under zero difference	0.05	17	11 / 6	0.807
Equivalence under zero difference	0.075	9	6 / 3	0.829
Equivalence under zero difference	0.1	7	4 / 3	0.856

Table S7. Exploratory binormal planning simulation for paired AUC differences. Two candidate AUCs are centred on 0.80, separated by the stated gap, with within-class score-noise correlation 0.5022 and healing allocation near 9/14. Each searched total uses 4,000 simulation draws and a model-based normal approximation for the paired difference at $\alpha=0.05$. Totals assume one observation per independent patient and no missing outcome. The gaps are hypothetical inputs; the approximate 80% power search does not establish actual clinical performance, calibrated patient-test power or a validated recruitment target.

Candidate AUC gap	Total observations	Healing / non-healing	Estimated power
0.05	721	464 / 257	0.818
0.1	181	116 / 65	0.800
0.15	84	54 / 30	0.816
0.2	52	33 / 19	0.811

Table S8. Association between mean fibroblast-associated topic loading within fibroblasts and the threshold-defined high-state fraction across 25 GSE165816 specimens. The Spearman row gives the observed correlation and percentile 95% limits from 10,000 sample-row bootstrap draws. The leave-one-out row gives the minimum and maximum of 25 recalculations, not a confidence interval; a dash means no single point estimate applies. A separate 10,000-draw permutation analysis gave a two-sided add-one probability of 0.00010 and a central null interval from -0.4025 to 0.4072. Samples may share patients, and the two summaries share cells.

Summary	Estimate	Lower endpoint	Upper endpoint
spearman rho	0.9325	0.7887	0.9828
leave one out rho	—	0.9235	0.9566

Table S9. External sample-level Spearman correlations with the frozen fibroblast fibroblast-associated topic mean under exclusive lineage gates. Immune fraction divides immune-assigned cells by all retained cells; B/plasma share divides B/plasma cells by immune-assigned cells. Detected genes and unique molecular identifiers (UMIs) are separate per-sample medians over QC cells. All entries are descriptive point estimates without confidence intervals. Specimens are not verified independent patients, and no listed cohort supplies a qualifying patient-level healing outcome.

Cohort	Samples	Immune fraction	B/plasma share	Detected genes	UMIs
GSE223964	8	-0.024	-0.071	-0.429	-0.476
GSE245703	12	0.846	0.077	-0.427	-0.308
GSE268834	8	0.714	-0.500	0.238	0.262

Table S10. Complete exploratory covariate screen in 25 discovery specimens. Each row correlates the fibroblast high-state fraction, using the historical rule loading at least 0.184, with one of 31 covariates. Lineage fractions use all retained cells; technical summaries and topic means use fibroblasts. Values are Spearman rho, two-sided asymptotic p and Benjamini–Hochberg q across all 31 tests, ordered by absolute rho; 14 rows have q below 0.05. UMI means unique molecular identifier. The G label is an identifier, not an established batch or time variable. Repeated patients, shared cells and compositional constraints limit independence; these marginal associations are not causal or healing-validation tests.

Covariate	Specimens	Spearman rho	Raw p	BH q
B/plasma fraction	25	0.8279	3.26e-07	1.01e-05
Fibroblast CFD/APOD mean	25	-0.7664	7.95e-06	0.000123
Fibroblast PTPRC-positive fraction	25	0.7267	3.88e-05	0.000401
Fibroblast median UMI count	25	0.6632	0.000302	0.00234
Fibroblast IGFBP7/TAGLN mean	25	0.6514	0.000421	0.00261
Sweat-gland fraction	25	-0.5832	0.00222	0.0114
Fibroblast FOS/IGFBP7 mean	25	-0.5687	0.00302	0.0134
Fibroblast DCN/CFD mean	25	-0.5592	0.00366	0.0142
Fibroblast IGFBP7/IGFBP5 mean	25	0.5316	0.00624	0.0215
Fibroblast COL1A1/COL1A2 mean	25	0.5238	0.00721	0.0223
Keratinocyte fraction	25	-0.4836	0.0143	0.0404
Fibroblast HLA-DRA/CD74 mean	25	0.4726	0.0171	0.0441
T-cell fraction	25	0.4647	0.0193	0.0459
Fibroblast TAGLN/ACTA2 mean	25	0.4552	0.0222	0.0492
Melanocyte fraction	25	-0.4442	0.0261	0.054
Fibroblast JUNB/DNAJB1 mean	25	0.4182	0.0375	0.0726
Mast-cell fraction	25	-0.3977	0.0489	0.0893
Myeloid fraction	25	0.3891	0.0546	0.094
Numeric G specimen label	25	-0.3607	0.0765	0.125
Endothelial fraction	25	-0.2347	0.259	0.401
Total retained cells	25	-0.2276	0.274	0.404
Fibroblast DNAJB1/FOS mean	25	0.1977	0.344	0.484
Fibroblast KRT14/KRT10 mean	25	0.1457	0.487	0.63
Fibroblast dissociation score	25	0.1449	0.489	0.63
Lymphatic endothelial fraction	25	-0.1331	0.526	0.63
Fibroblast fraction	25	0.1292	0.538	0.63
Fibroblast FOS/JUNB mean	25	-0.1260	0.548	0.63
Retained fibroblasts	25	0.0851	0.686	0.759
Pericyte/smooth-muscle fraction	25	-0.0473	0.823	0.879
Fibroblast IGFBP7/SPARCL1 mean	25	-0.0354	0.866	0.895
Fibroblast DCD/SAT1 mean	25	-0.0236	0.911	0.911

Table S11. Immune-transcript exclusion and residual-mixture controls. BIC denotes Bayesian information criterion, fitted to sample high-state fractions or their regression residuals; the difference is BIC for one Gaussian minus BIC for two. Positive differences favour two components. The first two rows share 24 matched specimens and have fraction correlation 0.9887. Residual regressions use all 25 specimens: immune covariates are B/plasma, T-cell and myeloid fractions plus immune-transcript positivity; placebos are pericyte/smooth-muscle, endothelial, melanocyte and sweat-gland fractions. R squared is variance explained by each regression and is inapplicable to unadjusted rows (dash). Both residual differences are negative, so the analysis does not establish immune-specific separation.

Summary	n	R squared	BIC one	BIC two	Difference
Before immune-transcript exclusion	24	—	12.417	-28.569	40.986
After immune-transcript exclusion	24	—	5.342	-51.341	56.683
Immune-covariate residuals	25	0.4953	-3.900	0.031	-3.931
Placebo-covariate residuals	25	0.3549	2.235	10.433	-8.198

Table S12. Historical sample-level lineage and loading contrasts. Ratios divide the healing-arm mean by the non-healing-arm mean; intervals are percentile 95% limits from 20,000 within-arm sample bootstraps, and p values are two-sided Mann–Whitney tests. GSE165816 uses 9 versus 5 DFU specimens; GSE231643 uses 5 versus 3 title-labelled specimens. The discovery ratios come from the saved lineage-projection analysis and are distinct from the later deterministic all-cell patient contrast. Fibroblast fraction uses all cells as denominator; fibroblast mean uses only assigned fibroblasts. These are separate estimands. Repeated discovery patients and unavailable external patient mappings prevent independent outcome validation.

Cohort	Readout	Mean ratio	Bootstrap 95% interval	Raw p
GSE165816	Fibroblast fraction	1.425	0.873 to 2.298	0.438
GSE165816	All-cell fibroblast-associated topic mean	2.724	1.369 to 5.732	0.0599
GSE165816	Fibroblast fibroblast-associated topic mean	2.889	1.059 to 13.966	0.042
GSE231643	Fibroblast fraction	2.051	1.128 to 4.184	0.0714
GSE231643	All-cell fibroblast-associated topic mean	2.564	1.310 to 8.079	0.143
GSE231643	Fibroblast fibroblast-associated topic mean	1.660	1.007 to 6.102	0.143

Table S13. All six paired AUC differences at each analysis unit. Each contrast subtracts the second leave-one-out readout from the first. Percentile 95% intervals use 4,000 common-index, arm-stratified bootstrap draws of fixed out-of-fold scores: 9 versus 5 specimens or 7 versus 4 mapped patients. PCA and NMF denote principal component analysis and nonnegative matrix factorization. No sample-unit interval excludes zero; patient-unit Fibroblast topic minus NMF and Module minus NMF intervals do. The intervals are exploratory, are not adjusted across pairs, and do not refit the entire representation in each bootstrap draw.

Unit	AUC contrast	Difference	Bootstrap 95% interval
14 specimens	Fibroblast topic minus Module	-0.0222	-0.2889 to 0.2222
14 specimens	Fibroblast topic minus PCA	0.2889	-0.0889 to 0.6667
14 specimens	Fibroblast topic minus NMF	-0.0556	-0.3556 to 0.2222
14 specimens	Module minus PCA	0.3111	0.0000 to 0.6667
14 specimens	Module minus NMF	-0.0333	-0.2889 to 0.2000
14 specimens	PCA minus NMF	-0.3444	-0.7111 to 0.0000
11 patients	Fibroblast topic minus Module	-0.0714	-0.4286 to 0.2857
11 patients	Fibroblast topic minus PCA	0.2143	-0.4643 to 0.9286
11 patients	Fibroblast topic minus NMF	0.5893	0.1429 to 0.9643
11 patients	Module minus PCA	0.2857	-0.1786 to 0.7857
11 patients	Module minus NMF	0.6607	0.2857 to 0.9652
11 patients	PCA minus NMF	0.3750	0.0000 to 0.7857

Table S14. Complete paired anatomical control across 15 topics in 10 author-mapped patients. Differences are mean all-cell loadings in foot minus forearm, after cell-count-weighted collapse of repeated foot specimens. Intervals are percentile 95% limits from 4,000 paired-patient bootstrap draws. Two-sided p values enumerate all 1,024 sign allocations and retain add-one correction; q values use Benjamini–Hochberg across 15 topics. The TIMP1/COL3A1, COL1A1/COL1A2, CFD/APOD and IGFBP7/TAGLN loadings have q below 0.05. Labels list the two leading decoder genes; they do not assert cell-type identity. Bootstrap intervals and permutation tests need not be dual. This is an anatomical sensitivity control within the discovery source, not a healing endpoint.

Leading genes	Foot minus forearm	Bootstrap 95% interval	Sign-flip p	BH q
TIMP1/COL3A1	0.0476	0.0119 to 0.0916	0.0107	0.0402
JUNB/DNAJB1	0.0103	-0.0616 to 0.0793	0.799	0.956
COL1A1/COL1A2	0.0914	0.0535 to 0.1376	0.00293	0.022
IGFBP7/IGFBP5	0.0188	0.0041 to 0.0361	0.0634	0.119
TAGLN/ACTA2	0.0016	-0.0200 to 0.0241	0.893	0.956
KRT14/KRT10	-0.2491	-0.4274 to -0.0682	0.0361	0.0774
CFD/APOD	-0.0362	-0.0567 to -0.0169	0.0107	0.0402
DCN/CFD	-0.0159	-0.0383 to 0.0042	0.208	0.312
DCD/SAT1	-0.0001	-0.0083 to 0.0098	0.975	0.975
IGFBP7/SPARCL1	0.0398	-0.0015 to 0.0897	0.134	0.223
FOS/IGFBP7	-0.0038	-0.0087 to 0.0025	0.245	0.334
HLA-DRA/CD74	0.0111	-0.0591 to 0.0984	0.854	0.956
IGFBP7/TAGLN	0.0715	0.0436 to 0.1039	0.00293	0.022
FOS/JUNB	0.0068	0.0025 to 0.0118	0.0185	0.0556
DNAJB1/FOS	0.0061	0.0020 to 0.0108	0.0224	0.0561

Table S15. Complete Welch two-one-sided-test sensitivity grids for the all-cell mean fibroblast-associated topic contrast. The symmetric margin is on the loading scale, and both lower and upper one-sided p values must be below 0.05 for the final column to be Yes. Sample rows use 9 versus 5 specimens; patient rows use 7 versus 4 mapped patients with cell-count-weighted repeat collapse. Margins were examined retrospectively and have no externally justified clinical meaning. A Yes entry therefore records a mathematical sensitivity result, not demonstrated clinical equivalence.

Unit	Margin	Lower-test p	Upper-test p	Both below 0.05
14 specimens	0.05	0.000143	0.359	No
14 specimens	0.10	3.54e-06	0.00438	Yes
14 specimens	0.20	2.07e-08	1.57e-06	Yes
14 specimens	0.30	6.17e-10	1.22e-08	Yes
14 specimens	0.50	5.18e-12	3.21e-11	Yes
11 patients	0.05	0.00131	0.13	No
11 patients	0.10	5.19e-05	0.00206	Yes
11 patients	0.20	7.01e-07	6e-06	Yes
11 patients	0.30	4.05e-08	1.77e-07	Yes
11 patients	0.50	9.05e-10	2.22e-09	Yes

Table S16. Patient-level robustness in the discovery cohort. The first row correlates fibroblast mean loading with the high-state fraction after weighting repeated specimen summaries by fibroblast count within 20 patients (25 specimens). Its percentile 95% interval uses 10,000 patient resamples, seed 217. The second row uses the previously defined all-cell healing contrast in 7 healing and 4 non-healing patients; its original interval remains in the outcome table. Omission ranges give the minimum and maximum after deleting one patient at a time (20 or 11 omissions), not confidence limits. Neither analysis refits the expression model, changes the state cut, adds patients, or provides external validation.

Quantity	Unit	Estimate	Patient bootstrap 95%	Patient-omission range
Mean versus high-state fraction	20 patients	0.9372	0.7760 to 0.9903	0.9266 to 0.9643
Healing minus non-healing	11 patients	0.0280	Not recalculated	0.0139 to 0.0347

Table S17. Topic-capacity sensitivity after tracking the TIMP1-carrying topic at each K. Gene-based descriptions identify the tracked loading; each fit supplies its own two-Gaussian midpoint cut. Semantic rho uses 25 specimens; the unadjusted healing-arm p value uses 9 versus 5 high-state-fraction summaries. The final column asks whether the maximum fraction occurs in healthy skin. At K=10 the tracked topic is a different IGFBP-rich programme and the counterexample fails. Topics across capacities are not assumed to have identical biological meaning.

K	Programme description	Derived cut	Semantic rho	Healing p	Healthy maximum
10	IGFBP-rich	0.1454	0.9469	0.147	No
15	CHI3L1/COL3A1-rich	0.1691	0.9316	0.109	Yes
20	CHI3L1/COL3A1-rich	0.1695	0.9236	0.109	Yes

Table S18. All 18 state-cut sensitivity evaluations for the frozen K=15 topic. Spearman rho correlates fibroblast mean loading with the threshold-defined high-state fraction across 25 specimens. The ratio compares mean high-state fractions in 9 healing versus 5 non-healing specimens; p is the unadjusted two-sided Mann–Whitney probability. The last column gives the group containing the largest fraction. The grid includes the fitted cut rounded to 0.1843. Four cuts have p below 0.05; these are dependent sensitivity analyses, not 18 independent confirmations.

State cut	Semantic rho	Healing / non-healing	Raw p	Largest fraction
0.0800	0.9326	2.966	0.042	Healthy
0.1000	0.9414	2.959	0.042	Healthy
0.1200	0.9091	3.011	0.042	Healthy
0.1400	0.9301	2.992	0.042	Healthy
0.1600	0.9294	2.981	0.0599	Healthy
0.1800	0.9349	3.001	0.0599	Healthy
0.1843	0.9325	2.975	0.0599	Healthy
0.2000	0.9349	2.987	0.0599	Healthy
0.2200	0.9341	3.124	0.0599	Healthy
0.2400	0.9228	3.352	0.0617	Healthy
0.2600	0.9228	3.424	0.0617	Healthy
0.2800	0.9236	3.465	0.0617	Healthy
0.3000	0.9236	3.499	0.0617	Healthy
0.3200	0.9479	3.741	0.0617	Healthy
0.3400	0.9463	3.699	0.0617	Healthy
0.3600	0.9401	4.040	0.0608	Healthy
0.3800	0.9182	3.945	0.0608	Healthy
0.4000	0.9182	3.889	0.0608	Healthy